

Multiple Choice

1. **Answer:** A

Options: A) Levels of oxygen, acidity or alkalinity of the medium; B) Temperature changes, light exposure; C) Atmospheric pressure, noise levels; D) Nitrogen levels, radiation exposure; E) Soil texture, light wavelength

Note: The growth of bacteria is significantly influenced by levels of oxygen, which determine whether the bacteria are aerobic or anaerobic, and the acidity or alkalinity of the medium, as these factors can affect enzyme activity and metabolic processes essential for bacterial proliferation.

2. **Answer:** C

Options: A) 50%; B) 100%; C) 0%; D) 10%; E) 75%

Note: The child cannot have type O blood because type O requires two recessive alleles (ii), while the individual with type AB blood has the genotype IAIB, which cannot provide an i allele for type O blood.

3. **Answer:** A

Options: A) Mate attraction, species recognition, defense, and territorial claim; B) For heat regulation; C) For hunting; D) For navigation; E) For mineral absorption

Note: The correct answer encompasses the essential biological functions of sound communication in animals, which facilitate reproduction through mate attraction, enhance survival through species recognition and defense mechanisms, and establish dominance and resource control through territorial claims.

4. **Answer:** C

Options: A) Ribosome; B) Mitochondrion; C) Chloroplast; D) Lysosome; E) Endoplasmic reticulum

Note: The targeting of a newly synthesized protein to the chloroplast requires two different signal peptides because proteins must first transit through the outer and inner membranes of the organelle, each requiring distinct targeting signals for proper localization.

5. **Answer:** D

Options: A) Alpha-ketoglutarate; B) Glutamate; C) Acetyl CoA; D) Oxaloacetate; E) Citrate

Note: Ornithine is regenerated in the urea cycle after each cycle, similar to how oxaloacetate is continuously replenished in the citric acid cycle, highlighting their roles as intermediates that facilitate ongoing metabolic processes.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Little is known about the mechanisms underlying irregular bleeding in HT users. This is the first report of uNK cells and their association with regulating cytokines in postmenopausal endometrium and demonstrates a possible mechanism by which HT may induce irregular bleeding.

7. **Answer:** True

Options: A) True; B) False

Note: There is good correlation and agreement between end-tidal CO(2) and arterial CO(2) in ELBW infants in the EtCO(2) range 30-50 mmHg. End-tidal CO(2) monitoring can be helpful in trending or for screening abnormal PaCO(2) values in ELBW infants in first week of life.

8. **Answer:** True

Options: A) True; B) False

Note: The study supports the hypothesis that nurses' aides who were exposed to ETS at home during childhood have an increased risk of long-term sickness absence.

9. **Answer:** True

Options: A) True; B) False

Note: For conditions with a well-developed quality of care evidence base, such as hypertension and diabetes, a single structured implicit review to assess the quality of care over a period of time is moderately reliable. This method could be a reasonable complement or alternative to explicit indicator approaches for assessing and comparing quality of care. Structured implicit review, like explicit quality measures, must be used more cautiously for illnesses for which the evidence base is less well developed, such as COPD and acute, short-course illnesses.

10. **Answer:** True

Options: A) True; B) False

Note: The presentation of the first twin at delivery can be determined by sonogram by the 32 nd week of gestation in over 90% of twin pregnancies.

Long Form Response

11. **Answer:** The theory of evolution is supported by various types of evidence from living organisms, each contributing uniquely to our understanding of evolutionary processes. Taxonomy reveals the hierarchical relationships between species, illustrating common ancestry through shared characteristics. Morphological comparisons highlight structural similarities and differences that indicate adaptive changes over time. Comparative biochemistry, particularly the analysis of DNA and protein sequences, provides molecular evidence of evolutionary relationships, showing how genetic information is conserved or modified across taxa. Embryological structures demonstrate

that many species share similar developmental pathways, suggesting a common origin. Lastly, advancements in genetics have unveiled the mechanisms of heredity and mutation, offering a framework for understanding evolutionary change at the molecular level. Together, these diverse lines of evidence create a robust and interconnected foundation for the theory of evolution, emphasizing the unity and diversity of life on Earth.

Note: correct because it identifies and elaborates on multiple lines of evidence-taxonomy, morphological comparisons, and comparative biochemistry-that collectively demonstrate the interconnectedness of species and their evolutionary relationships, thereby reinforcing the validity of the theory of evolution.

12. **Answer:** Leaching in tropical rain forest soils occurs when heavy rainfall washes away soluble nutrients, leading to nutrient depletion in the upper soil layers. This process is particularly pronounced in these ecosystems due to their high precipitation and warm temperatures, which facilitate rapid weathering of minerals. Nutrient cycling, however, is highly efficient in tropical rain forests, as rapid decomposition of organic matter by a diverse array of microorganisms and detritivores quickly replenishes nutrient availability, sustaining plant growth and biodiversity. The interplay between leaching and nutrient cycling profoundly influences ecosystem dynamics, as it determines soil fertility and, consequently, the diversity and productivity of plant communities. This intricate balance not only supports a rich tapestry of flora and fauna but also enhances the resilience of the ecosystem against disturbances, highlighting the importance of understanding these processes in conservation efforts.

Note: The answer accurately describes how leaching depletes nutrients from the upper soil layers in tropical rain forests due to heavy rainfall, while also highlighting the efficiency of nutrient cycling through rapid decomposition, which together influence ecosystem dynamics and biodiversity in this biome.